

USER INTERFACE DESIGN

EXPERIMENT 10: Create Data Visualization for Inventory Management System using JavaScript

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AIM:

To design a data visualization dashboard for an inventory management system using JavaScript, representing data through doughnut charts and line graphs.

PROCEDURE:

Step 1: Creating the HTML Structure

An HTML page was created containing:

- A main heading for the dashboard
- Two canvas elements for displaying charts
- Integration of the Chart.js library for visualization

Step 2: Styling the Interface using CSS

The user interface was enhanced with slight styling improvements:

- Gradient background for a modern appearance
- Center-aligned layout
- Card-style containers for charts
- Rounded corners and shadow effects
- Proper spacing and alignment

Step 3: Defining Inventory Data using JavaScript

Inventory data for different categories was defined:

- Mobiles – 180
- Laptops – 120
- Accessories – 90
- Furniture – 70
- Stationery – 40

Step 4: Creating Doughnut Chart

A Doughnut Chart was generated using Chart.js to display the proportional distribution of inventory across categories.

Step 5: Creating Line Graph

A Line Graph was created using Chart.js to represent stock levels across different categories and visualize trends

.

CODE:

INDEX.HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">

<title>Inventory Dashboard</title>

<script
src="https://cdn.jsdelivr.net/npm/chart.js@4.4.0/dist/chart.umd.j
s"></script>

<style>
body{
```

```
    font-family: Arial, sans-serif;
    margin:0;
    padding:20px;
    background: linear-gradient(to right, #1e3c72, #2a5298);
    color:white;
}

h1{
    text-align:center;
    margin-bottom:30px;
}

.container{
    display:flex;
    flex-wrap:wrap;
    justify-content:center;
    gap:30px;
}

.chart-card{
    background:white;
    color:#333;
    padding:20px;
    border-radius:15px;
    box-shadow:0 8px 20px rgba(0,0,0,0.3);
    width:500px;
}

canvas{
    width:100% !important;
    height:auto !important;
}
</style>

</head>
```

```

<body>

<h1>Inventory Analytics Dashboard</h1>

<div class="container">

  <div class="chart-card">
    <h3>Category Distribution (Doughnut Chart)</h3>
    <canvas id="doughnutChart"></canvas>
  </div>

  <div class="chart-card">
    <h3>Stock Trend Overview (Line Chart)</h3>
    <canvas id="lineChart"></canvas>
  </div>

</div>

<script src="script.js" defer></script>

</body>
</html>

```

SCRIPT.JS:

```

const labels =
['Mobiles', 'Laptops', 'Accessories', 'Furniture', 'Stationery'];

const dataValues = [180, 120, 90, 70, 40];

// Doughnut Chart
new Chart(document.getElementById('doughnutChart'), {
  type: 'doughnut',
  data: {
    labels: labels,
    datasets: [{
      label: 'Stock Distribution',

```

```

        data: dataValues,
        backgroundColor: [
            '#ff6384',
            '#36a2eb',
            '#ffce56',
            '#4bc0c0',
            '#9966ff'
        ]
    }
}
});

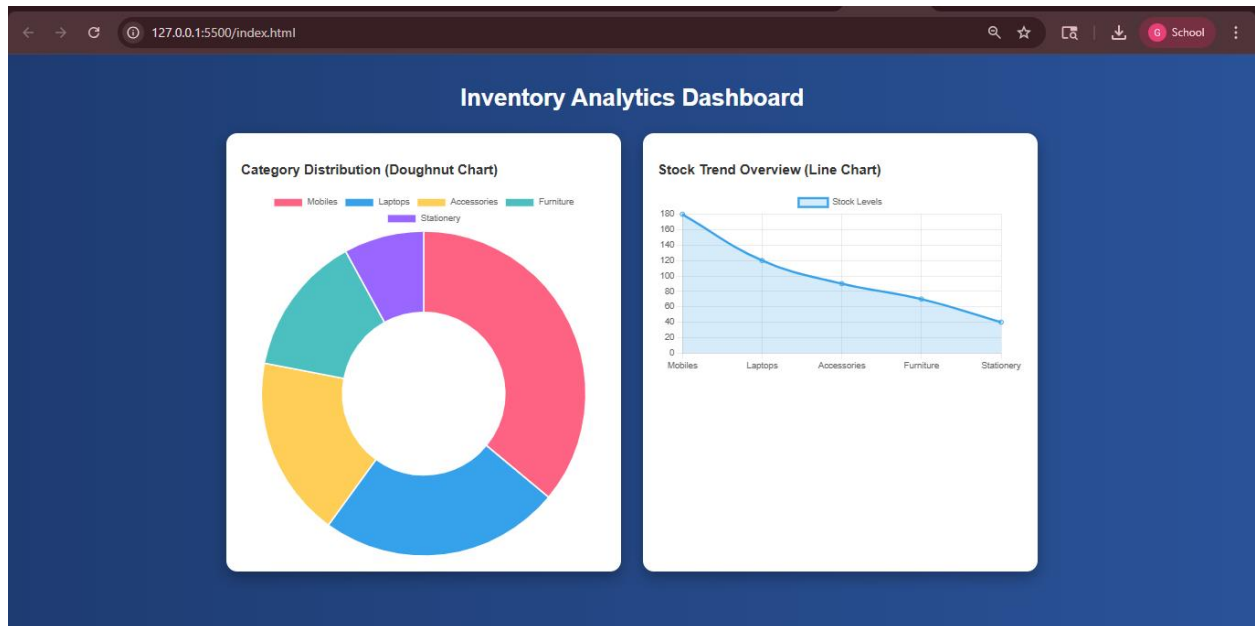
// Line Chart
new Chart(document.getElementById('lineChart'), {
    type: 'line',
    data: {
        labels: labels,
        datasets: [{
            label: 'Stock Levels',
            data: dataValues,
            borderColor: '#36a2eb',
            backgroundColor: 'rgba(54,162,235,0.2)',
            fill: true,
            tension: 0.4
        }]
    },
    options: {
        scales: {
            y: {
                beginAtZero: true
            }
        }
    }
});

```

OUTPUT:

The Inventory Management dashboard was created successfully.

- Pie Chart displays stock distribution by category.
- Bar Graph displays items in stock by category.
- Attractive user interface with charts was generated successfully.



PROTOTYPE LINK:

<http://127.0.0.1:5500/index.html>

RESULT:

The inventory dashboard was successfully created.

- The Doughnut Chart displays category-wise distribution effectively
- The Line Graph represents stock levels clearly
- The user interface is visually enhanced with improved styling
- The system provides an interactive and clear representation of inventory data